

PROJECT ATLAS

ATLAS-DES-002: SOFTWARE ARCHITECTURE SPECIFICATION

1. Architectural Pattern: The Central Truth

Atlas is built on a **Mediated Observer Pattern**. The system state is centralized. Views (UI/LED Display) do not share information directly; they synchronize against a single source of truth: the `AtlasStateEngine`.

2. Module Decomposition

2.1 `src.core.engine (AtlasStateEngine)`

Responsibility: Final authority on system state. Implements the State Machine (Startup -> Ready -> Armed -> Running -> Finished).

2.2 `src.gui.operator (AtlasOperatorUI)`

Responsibility: Operator interaction. Captures "Next Heat" parameters and sends "Transition Requests" to the Engine.

2.3 `src.gui.renderer (AtlasPublicDisplay)`

Responsibility: Painting the P3/P10 LED Matrix. A borderless PySide6 window rendering high-contrast timing data.

2.4 src.core.timers (AtlasTimerEngine)

Responsibility: The system heartbeat. Runs on a dedicated QThread or High-Precision Timer to ensure zero-drift countdowns.

3. State Transition Matrix

Current State	Event	Next State	Authorized By
READY	Heat Validated	ARMED	Engine
ARMED	Activate Pressed	RUNNING	Engine
RUNNING	Timer Expired	FINISHED	Timer Thread
ANY	Master Cancel	CANCELLED	Safety Override

4. Implementation Strategy

Atlas will use **PySide6 Signals and Slots** for inter-module communication. This ensures thread safety when the Timer Engine updates the Renderer without blocking the Operator UI.